

Lecture 11: Properties of OLS in multiple regression

Economics 326 — Introduction to Econometrics II

Vadim Marmer, UBC

Multiple regression and OLS

- Consider the multiple regression model with k regressors:

$$Y_i = \beta_0 + \beta_1 X_{1,i} + \beta_2 X_{2,i} + \dots + \beta_k X_{k,i} + U_i.$$

- Let $\hat{\beta}_0, \hat{\beta}_1, \dots, \hat{\beta}_k$ be the OLS estimators: if

$$\hat{U}_i = Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_{1,i} - \hat{\beta}_2 X_{2,i} - \dots - \hat{\beta}_k X_{k,i},$$

then

$$\sum_{i=1}^n \hat{U}_i = \sum_{i=1}^n X_{1,i} \hat{U}_i = \dots = \sum_{i=1}^n X_{k,i} \hat{U}_i = 0.$$

Multiple regression and OLS

- As in Lecture 9, we can write $\hat{\beta}_1$ as

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n \tilde{X}_{1,i} Y_i}{\sum_{i=1}^n \tilde{X}_{1,i}^2}, \text{ where}$$

– $\tilde{X}_{1,i}$ are the fitted OLS residuals: $\tilde{X}_{1,i} = X_{1,i} - \hat{\gamma}_0 - \hat{\gamma}_2 X_{2,i} - \dots - \hat{\gamma}_k X_{k,i}$.

– $\hat{\gamma}_0, \hat{\gamma}_2, \dots, \hat{\gamma}_k$ are the OLS coefficients: $\sum_{i=1}^n \tilde{X}_{1,i} = \sum_{i=1}^n \tilde{X}_{1,i} X_{2,i} = \dots = \sum_{i=1}^n \tilde{X}_{1,i} X_{k,i} = 0$.

- Similarly, we can write $\hat{\beta}_2$ as

$$\hat{\beta}_2 = \frac{\sum_{i=1}^n \tilde{X}_{2,i} Y_i}{\sum_{i=1}^n \tilde{X}_{2,i}^2}, \text{ where}$$

– $\tilde{X}_{2,i}$ are the fitted OLS residuals: $\tilde{X}_{2,i} = X_{2,i} - \hat{\delta}_0 - \hat{\delta}_1 X_{1,i} - \hat{\delta}_3 X_{3,i} - \dots - \hat{\delta}_k X_{k,i}$.

– $\hat{\delta}_0, \hat{\delta}_1, \hat{\delta}_3, \dots, \hat{\delta}_k$ are the OLS coefficients: $\sum_{i=1}^n \tilde{X}_{2,i} = \sum_{i=1}^n \tilde{X}_{2,i} X_{1,i} = \sum_{i=1}^n \tilde{X}_{2,i} X_{3,i} = \dots = \sum_{i=1}^n \tilde{X}_{2,i} X_{k,i} = 0$.

The OLS estimators are linear

- Consider $\hat{\beta}_1$:

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n \tilde{X}_{1,i} Y_i}{\sum_{i=1}^n \tilde{X}_{1,i}^2} = \sum_{i=1}^n \frac{\tilde{X}_{1,i}}{\sum_{l=1}^n \tilde{X}_{1,l}^2} Y_i = \sum_{i=1}^n w_{1,i} Y_i,$$

where

$$w_{1,i} = \frac{\tilde{X}_{1,i}}{\sum_{l=1}^n \tilde{X}_{1,l}^2}.$$

- Recall that $\tilde{X}_1$ are the residuals from a regression of X_1 against $X_2, \dots, X_k$ and a constant, and therefore $w_{1,i}$ depends only on $\mathbf{X}$.

Unbiasedness

- Suppose that
 - $Y_i = \beta_0 + \beta_1 X_{1,i} + \beta_2 X_{2,i} + \dots + \beta_k X_{k,i} + U_i$.
 - Conditional on $\mathbf{X}$** , $E[U_i | \mathbf{X}] = 0$ for all i 's.
 - Conditioning on $\mathbf{X}$ means that we condition on all regressors for all observations: $\mathbf{X} = \{(X_{1,i}, X_{2,i}, \dots, X_{k,i}) : i = 1, \dots, n\}$.
- Under the above assumptions, conditional on $\mathbf{X}$:

$$E[\hat{\beta}_0 | \mathbf{X}] = \beta_0,$$

$$E[\hat{\beta}_1 | \mathbf{X}] = \beta_1,$$

$\vdots$

$$E[\hat{\beta}_k | \mathbf{X}] = \beta_k.$$

Proof of unbiasedness

- Substituting Y_i and expanding:

$$\begin{aligned} \hat{\beta}_1 &= \frac{\sum_{i=1}^n \tilde{X}_{1,i} Y_i}{\sum_{i=1}^n \tilde{X}_{1,i}^2} = \frac{\sum_{i=1}^n \tilde{X}_{1,i} (\beta_0 + \beta_1 X_{1,i} + \beta_2 X_{2,i} + \dots + \beta_k X_{k,i} + U_i)}{\sum_{i=1}^n \tilde{X}_{1,i}^2} \\ &= \beta_0 \frac{\sum_{i=1}^n \tilde{X}_{1,i}}{\sum_{i=1}^n \tilde{X}_{1,i}^2} + \beta_1 \frac{\sum_{i=1}^n \tilde{X}_{1,i} X_{1,i}}{\sum_{i=1}^n \tilde{X}_{1,i}^2} + \beta_2 \frac{\sum_{i=1}^n \tilde{X}_{1,i} X_{2,i}}{\sum_{i=1}^n \tilde{X}_{1,i}^2} \\ &\quad + \dots + \beta_k \frac{\sum_{i=1}^n \tilde{X}_{1,i} X_{k,i}}{\sum_{i=1}^n \tilde{X}_{1,i}^2} + \frac{\sum_{i=1}^n \tilde{X}_{1,i} U_i}{\sum_{i=1}^n \tilde{X}_{1,i}^2}. \end{aligned}$$

- Using the partitioned regression results from Lecture 9:

$$\begin{aligned} \sum_{i=1}^n \tilde{X}_{1,i} &= \sum_{i=1}^n \tilde{X}_{1,i} X_{2,i} = \dots = \sum_{i=1}^n \tilde{X}_{1,i} X_{k,i} = 0, \\ \sum_{i=1}^n \tilde{X}_{1,i} X_{1,i} &= \sum_{i=1}^n \tilde{X}_{1,i}^2. \end{aligned}$$

- Therefore,

$$\hat{\beta}_1 = \beta_1 + \frac{\sum_{i=1}^n \tilde{X}_{1,i} U_i}{\sum_{i=1}^n \tilde{X}_{1,i}^2}.$$

Proof of unbiasedness

- We have

$$\hat{\beta}_1 = \beta_1 + \frac{\sum_{i=1}^n \tilde{X}_{1,i} U_i}{\sum_{i=1}^n \tilde{X}_{1,i}^2}.$$

- **Conditional on $\mathbf{X}$,**

$$E[U_i | \mathbf{X}] = 0.$$

- Therefore, **conditional on $\mathbf{X}$,**

$$\begin{aligned} E[\hat{\beta}_1 | \mathbf{X}] &= E\left[\beta_1 + \frac{\sum_{i=1}^n \tilde{X}_{1,i} U_i}{\sum_{i=1}^n \tilde{X}_{1,i}^2} \mid \mathbf{X}\right] \\ &= \beta_1 + \frac{\sum_{i=1}^n \tilde{X}_{1,i} E[U_i | \mathbf{X}]}{\sum_{i=1}^n \tilde{X}_{1,i}^2} \\ &= \beta_1. \end{aligned}$$

Conditional variance of the OLS estimators

- Suppose that:
 1. $Y_i = \beta_0 + \beta_1 X_{1,i} + \beta_2 X_{2,i} + \dots + \beta_k X_{k,i} + U_i$.
 2. **Conditional on $\mathbf{X}$,** $E[U_i | \mathbf{X}] = 0$ for all i 's.
 3. **Conditional on $\mathbf{X}$,** $E[U_i^2 | \mathbf{X}] = \sigma^2$ for all i 's.
 4. **Conditional on $\mathbf{X}$,** $E[U_i U_j | \mathbf{X}] = 0$ for all $i \neq j$.
- The conditional variance of $\hat{\beta}_1$ given $\mathbf{X}$ is

$$\text{Var}(\hat{\beta}_1 | \mathbf{X}) = \frac{\sigma^2}{\sum_{i=1}^n \tilde{X}_{1,i}^2}.$$

- **Gauss-Markov Theorem:** Under Assumptions 1–4, the OLS estimators are **BLUE**.

Derivation of the conditional variance

- We have $\hat{\beta}_1 = \beta_1 + \frac{\sum_{i=1}^n \tilde{X}_{1,i} U_i}{\sum_{i=1}^n \tilde{X}_{1,i}^2}$.

- Conditional on $\mathbf{X}$,

$$\begin{aligned}
\text{Var}(\hat{\beta}_1 | \mathbf{X}) &= \text{E} \left[\left(\hat{\beta}_1 - \text{E}[\hat{\beta}_1 | \mathbf{X}] \right)^2 | \mathbf{X} \right] \\
&= \text{E} \left[\left(\frac{\sum_{i=1}^n \tilde{X}_{1,i} U_i}{\sum_{i=1}^n \tilde{X}_{1,i}^2} \right)^2 | \mathbf{X} \right] \\
&= \frac{1}{\left(\sum_{i=1}^n \tilde{X}_{1,i}^2 \right)^2} \text{E} \left[\left(\sum_{i=1}^n \tilde{X}_{1,i} U_i \right)^2 | \mathbf{X} \right] \\
&= \frac{1}{\left(\sum_{i=1}^n \tilde{X}_{1,i}^2 \right)^2} \left(\sum_{i=1}^n \tilde{X}_{1,i}^2 \sigma^2 + \sum_{i \neq j} \tilde{X}_{1,i} \tilde{X}_{1,j} \cdot 0 \right) \\
&= \frac{\sigma^2 \sum_{i=1}^n \tilde{X}_{1,i}^2}{\left(\sum_{i=1}^n \tilde{X}_{1,i}^2 \right)^2} = \frac{\sigma^2}{\sum_{i=1}^n \tilde{X}_{1,i}^2}.
\end{aligned}$$

Conditional covariance of the OLS estimators

- Consider $\hat{\beta}_1$ and $\hat{\beta}_2$:

$$\begin{aligned}
\hat{\beta}_1 &= \beta_1 + \frac{\sum_{i=1}^n \tilde{X}_{1,i} U_i}{\sum_{i=1}^n \tilde{X}_{1,i}^2}, \\
\hat{\beta}_2 &= \beta_2 + \frac{\sum_{i=1}^n \tilde{X}_{2,i} U_i}{\sum_{i=1}^n \tilde{X}_{2,i}^2},
\end{aligned}$$

where

- $\tilde{X}_1$ are the fitted residuals from the regression of X_1 against a constant and $X_2, X_3, \dots, X_k$.
- $\tilde{X}_2$ are the fitted residuals from the regression of X_2 against a constant and $X_1, X_3, \dots, X_k$.

- We will show that given Assumptions 1–4, **conditional on $\mathbf{X}$** :

$$\text{Cov}(\hat{\beta}_1, \hat{\beta}_2 | \mathbf{X}) = \sigma^2 \frac{\sum_{i=1}^n \tilde{X}_{1,i} \tilde{X}_{2,i}}{\sum_{i=1}^n \tilde{X}_{1,i}^2 \sum_{i=1}^n \tilde{X}_{2,i}^2}$$

Conditional covariance of the OLS estimators

Conditional on $\mathbf{X}$,

$$\begin{aligned}
&\text{Cov}(\hat{\beta}_1, \hat{\beta}_2 | \mathbf{X}) \\
&= \text{E} \left[\left(\hat{\beta}_1 - \text{E}[\hat{\beta}_1 | \mathbf{X}] \right) \left(\hat{\beta}_2 - \text{E}[\hat{\beta}_2 | \mathbf{X}] \right) | \mathbf{X} \right] \\
&= \frac{1}{\sum_{i=1}^n \tilde{X}_{1,i}^2 \sum_{i=1}^n \tilde{X}_{2,i}^2} \text{E} \left[\left(\sum_{i=1}^n \tilde{X}_{1,i} U_i \right) \left(\sum_{i=1}^n \tilde{X}_{2,i} U_i \right) | \mathbf{X} \right] \\
&= \frac{1}{\sum_{i=1}^n \tilde{X}_{1,i}^2 \sum_{i=1}^n \tilde{X}_{2,i}^2} \left(\sum_{i=1}^n \tilde{X}_{1,i} \tilde{X}_{2,i} \sigma^2 + \sum_{i \neq j} \tilde{X}_{1,i} \tilde{X}_{2,j} \cdot 0 \right) \\
&= \sigma^2 \frac{\sum_{i=1}^n \tilde{X}_{1,i} \tilde{X}_{2,i}}{\sum_{i=1}^n \tilde{X}_{1,i}^2 \sum_{i=1}^n \tilde{X}_{2,i}^2}.
\end{aligned}$$

Normality of the OLS estimators

- In addition to Assumptions 1–4, assume that **conditional on $\mathbf{X}$** , U_i 's are jointly normally distributed.
- $\hat{\beta}_0, \hat{\beta}_1, \dots, \hat{\beta}_k$ are linear estimators:

$$\hat{\beta}_j = \sum_{i=1}^n w_{j,i} Y_i = \beta_j + \sum_{i=1}^n w_{j,i} U_i,$$

where

$$w_{j,i} = \frac{\tilde{X}_{j,i}}{\sum_{l=1}^n \tilde{X}_{j,l}^2},$$

and $\tilde{X}_{j,i}$ are the residuals from the regression of X_j against the rest of the regressors.

- It follows that $\hat{\beta}_0, \hat{\beta}_1, \dots, \hat{\beta}_k$ are jointly normally distributed (conditional on $\mathbf{X}$).

Inclusion of irrelevant regressors

- Suppose that the true model is $Y_i = \beta_0 + \beta_1 X_{1,i} + U_i$.
- We could estimate β_1 by

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (X_{1,i} - \bar{X}_1) Y_i}{\sum_{i=1}^n (X_{1,i} - \bar{X}_1)^2}.$$

- Suppose that instead we regress Y against a constant, X_1 and additional $k - 1$ regressors $X_2, \dots, X_k$, i.e., we estimate β_1 by

$$\tilde{\beta}_1 = \frac{\sum_{i=1}^n \tilde{X}_{1,i} Y_i}{\sum_{i=1}^n \tilde{X}_{1,i}^2}.$$

- We have

$$\begin{aligned} \tilde{\beta}_1 &= \frac{\sum_{i=1}^n \tilde{X}_{1,i} (\beta_0 + \beta_1 X_{1,i} + U_i)}{\sum_{i=1}^n \tilde{X}_{1,i}^2} \\ &= \beta_1 + \frac{\sum_{i=1}^n \tilde{X}_{1,i} U_i}{\sum_{i=1}^n \tilde{X}_{1,i}^2}. \end{aligned}$$

- Since conditional on $\mathbf{X}$, $E[U_i | \mathbf{X}] = 0$, $\tilde{\beta}_1$ is **unbiased**!

Inclusion of irrelevant regressors

- When $Y_i = \beta_0 + \beta_1 X_{1,i} + U_i$,

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (X_{1,i} - \bar{X}_1) Y_i}{\sum_{i=1}^n (X_{1,i} - \bar{X}_1)^2} \quad \text{and}$$

$$\tilde{\beta}_1 = \frac{\sum_{i=1}^n \tilde{X}_{1,i} Y_i}{\sum_{i=1}^n \tilde{X}_{1,i}^2}$$

are both unbiased.

- Conditional on $\mathbf{X}$,

$$\text{Var}(\hat{\beta}_1 | \mathbf{X}) = \frac{\sigma^2}{\sum_{i=1}^n (X_{1,i} - \bar{X}_1)^2} \quad \text{and}$$

$$\text{Var}(\tilde{\beta}_1 | \mathbf{X}) = \frac{\sigma^2}{\sum_{i=1}^n \tilde{X}_{1,i}^2}.$$

- Since the true model has only X_1 , by the Gauss-Markov Theorem $\hat{\beta}_1$ is BLUE and

$$\text{Var}(\hat{\beta}_1 | \mathbf{X}) \leq \text{Var}(\tilde{\beta}_1 | \mathbf{X}).$$

- Without the Gauss-Markov Theorem, one can show directly that $\sum_{i=1}^n (X_{1,i} - \bar{X}_1)^2 \geq \sum_{i=1}^n \tilde{X}_{1,i}^2$.

Proof of the variance inequality

- $\tilde{X}_{1,i}$ are the fitted residuals from regressing $X_{1,i}$ against a constant, $X_{2,i}, \dots, X_{k,i}$:

$$X_{1,i} = \hat{\gamma}_0 + \hat{\gamma}_2 X_{2,i} + \dots + \hat{\gamma}_k X_{k,i} + \tilde{X}_{1,i}.$$

- Consider the sums-of-squares for this regression:

$$SST_1 = \sum_{i=1}^n (X_{1,i} - \bar{X}_1)^2,$$

$$SSE_1 = \sum_{i=1}^n (\hat{\gamma}_0 + \hat{\gamma}_2 X_{2,i} + \dots + \hat{\gamma}_k X_{k,i} - \bar{X}_1)^2,$$

$$SSR_1 = \sum_{i=1}^n \tilde{X}_{1,i}^2.$$

- Thus,

$$\sum_{i=1}^n (X_{1,i} - \bar{X}_1)^2 - \sum_{i=1}^n \tilde{X}_{1,i}^2 = SST_1 - SSR_1 = SSE_1 \geq 0.$$

Variance and the number of regressors

- In $Y_i = \hat{\beta}_0 + \hat{\beta}_1 X_{1,i} + \hat{\beta}_2 X_{2,i} + \dots + \hat{\beta}_k X_{k,i} + \hat{U}_i$, the variance of the OLS estimator $\hat{\beta}_1$ is

$$\text{Var}(\hat{\beta}_1 | \mathbf{X}) = \frac{\sigma^2}{\sum_{i=1}^n \tilde{X}_{1,i}^2} = \frac{\sigma^2}{SSR_1},$$

where SSR_1 is the residual sum-of-squares from the regression of X_1 against a constant and the rest of the regressors.

- Since SSR_1 can only decrease when we add more regressors, $\text{Var}(\hat{\beta}_1 | \mathbf{X})$ increases with k , if the added regressors are irrelevant but correlated with the included regressors.
- If the added regressors are uncorrelated with X_1 , inclusion of such regressors will not affect SSR_1 (in large samples) or the variance of $\hat{\beta}_1$.
- If the added regressors are uncorrelated with X_1 and affect Y , their inclusion will reduce σ^2 without affecting SSR_1 and will reduce the variance of $\hat{\beta}_1$.

Estimation of variances and covariances

- In $Y_i = \hat{\beta}_0 + \hat{\beta}_1 X_{1,i} + \hat{\beta}_2 X_{2,i} + \dots + \hat{\beta}_k X_{k,i} + \hat{U}_i$,

$$\begin{aligned} \text{Var}(\hat{\beta}_1 | \mathbf{X}) &= \frac{\sigma^2}{\sum_{i=1}^n \tilde{X}_{1,i}^2}, \\ \text{Cov}(\hat{\beta}_1, \hat{\beta}_2 | \mathbf{X}) &= \sigma^2 \frac{\sum_{i=1}^n \tilde{X}_{1,i} \tilde{X}_{2,i}}{\sum_{i=1}^n \tilde{X}_{1,i}^2 \sum_{i=1}^n \tilde{X}_{2,i}^2}. \end{aligned}$$

- Variances and covariances can be estimated by replacing σ^2 with

$$s^2 = \frac{1}{n - k - 1} \sum_{i=1}^n \hat{U}_i^2.$$

- Estimated variance and covariance:

$$\begin{aligned} \widehat{\text{Var}}(\hat{\beta}_1) &= \frac{s^2}{\sum_{i=1}^n \tilde{X}_{1,i}^2}, \\ \widehat{\text{Cov}}(\hat{\beta}_1, \hat{\beta}_2) &= s^2 \frac{\sum_{i=1}^n \tilde{X}_{1,i} \tilde{X}_{2,i}}{\sum_{i=1}^n \tilde{X}_{1,i}^2 \sum_{i=1}^n \tilde{X}_{2,i}^2}. \end{aligned}$$